

# Customer Shopping Behavior Analysis

A Comprehensive Study of 3,900 Transaction Records Mapping Demographics, Purchasing Habits, and Subscription Loyalty Frameworks.



01 / EXECUTIVE SCOPE

# Unveiling Consumer Shopping Behavior

This project analyzes consumer behavior using high-fidelity transactional data across **3,900 distinct purchases**.

Our core initiative is uncovering deep patterns in customer segments, seasonal product preferences, spending tiers, and subscription behaviors to formulate concrete, data-backed strategic decisions.





# Data Architecture Summary

## Core Dataset Metrics

A clean tabular landscape comprising **3,900 rows** and **18 relational features**. Clean parsing yielded zero extreme skewness, with high-integrity metrics across features.

**Demographics:** Age, Gender, Geography, Subscriptions

**Transactions:** Item Categories, Amount, Seasons, Colors

**Analytics:** Discounts, Previous Volume, Review Ratings

## Cleanliness & Preprocessing

- ✓ **Missing Data Resolved:** Imputed 37 missing ratings via category medians.
- ✓ **Clean Formatting:** Standardized features into snake\_case format.
- ✓ **Redundancy Control:** Identified overlap and removed promo codes.
- ✓ **System Integration:** Connected Python pipeline directly to PostgreSQL.



# Revenue Distribution by Gender



## Demographic Revenue Breakdown

Our financial summary shows that **Male** spending drives the vast majority of transactions, contributing more than double the revenue generated by the **Female** category.

|                        |           |
|------------------------|-----------|
| ● Male Spend (67.7%)   | \$157,890 |
| ● Female Spend (32.3%) | \$75,191  |



# Subscription Performance Analysis

## Leveraging Loyalty Systems

Subscribers and Non-subscribers exhibit highly consistent spending baselines, with average purchase figures sitting within cents of each other.

This shows that while subscription models do not raise the ticket size on individual transactions, they are foundational for predictable, recurring revenue flow.

| STATUS         | CUSTOMERS | AVG SPEND | TOTAL REVENUE |
|----------------|-----------|-----------|---------------|
| Subscribed     | 1,053     | \$59.49   | \$62,645      |
| Non-Subscribed | 2,847     | \$59.87   | \$170,436     |



# Shipping Type & Value Matrix

**\$60.48**

EXPRESS SHIPPING AVG

**\$58.46**

STANDARD SHIPPING AVG

## Shipping Preferences & Basket Values

Comparing shipping metrics reveals an interesting consumer behavior pattern:  
**Express shipping users do not present significantly larger baskets.**

-  Minimal friction between standard and fast shipping tiers (\$2.02 variation).
-  Delivery choice is driven by immediate utility and convenience, not higher purchase capacity.



# Customer Loyalty Framework



## Loyal

**3,116 Customers**

The core foundation of active buyers with high transaction volume and highly predictable purchasing habits.



## Returning

**701 Customers**

Moderate engagement history. Highly responsive to customized, season-specific push notifications.



## New

**83 Customers**

Newly acquired users representing fresh retention opportunities via onboarding flows.



# Top Products by Category

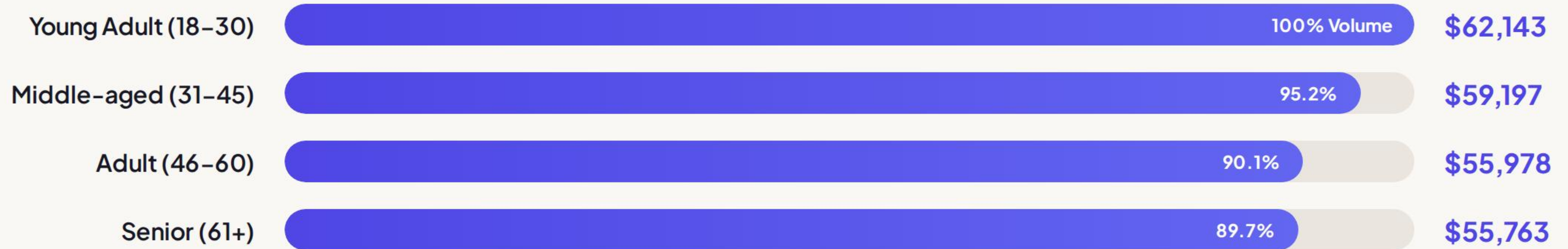


## Product Category Breakdown

| CATEGORY    | RANK #1 | RANK #2    | RANK #3  |
|-------------|---------|------------|----------|
| Accessories | Jewelry | Sunglasses | Belt     |
| Clothing    | Blouse  | Pants      | Shirt    |
| Footwear    | Sandals | Shoes      | Sneakers |
| Outerwear   | Jacket  | Coat       | -        |



# Revenue by Generational Segment



*Note: Revenue is evenly distributed across age demographics, with Young Adults leading by a small margin.*



# Rating and Discount Vulnerability

## ★ Top Products by Rating

| ITEM    | AVG RATING |
|---------|------------|
| Gloves  | 3.86 / 5.0 |
| Sandals | 3.84 / 5.0 |
| Boots   | 3.82 / 5.0 |

## 📉 Discount-Dependent Items

| ITEM     | DISCOUNT SHARE |
|----------|----------------|
| Hat      | 50.00%         |
| Sneakers | 49.66%         |
| Coat     | 49.07%         |



# Executive Behavior Dashboard

## Power BI Integration Metrics

A streamlined Business Intelligence dashboard connects our cleaned database to present transactional data visually.

This interface provides direct filtering capabilities across subscription tiers, geographic hotspots, and shipping methods, facilitating real-time strategic review.

**3.9K**  
CUSTOMERS

**\$59.76**  
AVG SPEND

**3.75**  
AVG RATING





# Tactical Business Recommendations

## Enhance Subscription Benefits

Drive value through exclusive subscriber benefits to increase volume, as current average spend is uniform across groups.

## Loyalty Program Upgrades

Incentivize "Returning" customers to enter the "Loyal" group using personalized trigger campaigns.

## Review Discounting Policy

Protect margin on discount-dependent categories (e.g., Hats at 50% discount rate) by bundling with high-margin jewelry items.

## Targeted Demographics

Align marketing spend with higher revenue groups, specifically optimizing outreach toward the highly active Young Adult cohort.



# Image Sources



[https://static.vecteezy.com/system/resources/previews/055/775/984/non\\_2x/shopping-bag-display-urban-marketplace-product-photography-minimalist-design-close-up-view-modern-aesthetics-png.png](https://static.vecteezy.com/system/resources/previews/055/775/984/non_2x/shopping-bag-display-urban-marketplace-product-photography-minimalist-design-close-up-view-modern-aesthetics-png.png)

Source: [www.vecteezy.com](https://www.vecteezy.com)

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Source: [www.vecteezy.com](https://www.vecteezy.com)

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